

FEDERATED WIND SPEED FORECASTING IN FINLAND: GEOGRAPHIC VS. CORRELATION GRAPHS WITH SYNCHRONOUS FEDGD

ABSTRACT

The goal of this report is to investigate whether federated learning (FL) over a weather station graph improves next-hour wind speed forecasting. The report uses completely independent local models as the baseline and compares them to the FL model using the weather station graphs. As the dataset, the report uses the Finnish Meteorological Institute (FMI) [1] hourly observation data from 2024.

For the FL models, the report constructs two graph topologies and compares them to the baseline model. The first graph topology is geographic k-NN which creates k edges between the k geographically closest nodes. The second topology is the correlation k-NN. In this graph, nodes have edges between their k most highly correlated (similar) neighbors. Then the FL systems are trained with FedGD [2] and compared with each other.

After training the baseline and both FL models, the report finds that both FL systems achieve negligible improvement over the baseline, with the α -sweep selecting a small regularization strength ($\alpha = 0.1$), indicating that strong collaboration is harmful. The result is explained by the local model being already well-specified. The local model has an abundance of data (one year) and a simple linear model leaves little room for graph regularization to improve forecasting.

Index Terms— Federated learning, networks, personalised machine learning, trustworthy artificial intelligence

1. INTRODUCTION

Wind speed forecasting is valuable for grid operators managing Finland’s growing wind energy capacity. Wind energy consumption has been stably growing from year 2015 and in year 2024, the share of Finland energy consumption that was produced with wind energy was 24% [6].

The Finnish Meteorological Institute (FMI) provides extensive data on their station network publicly. They operate a dense network of stations across Finland and this report tests the hypothesis of whether a collaborative federated learning (FL) network might help create better wind speed predictions compared to local models.

Using federated learning is relevant for these types of weather prediction problems because raw station data might be commercially sensitive or subject to data-residency con-

straints thus ruling out centralized training. Another reason to use federated learning is if the data is sparse or the sensors are prone to making mistakes, local models can borrow predictive power from neighboring stations to improve their own predictions. While personalized FL has been applied to wind power forecasting over small wind farm networks [3], graph-regularized approaches that exploit the spatial structure of dense public meteorological networks for wind speed are still fairly unexplored.

The objective of this report is to create a federated model that predicts the next-hour wind speed from the three previous hours. The report trains a local linear model and then creates a FL graph that encodes which stations should influence each other during training. The key design question in this report is whether the choice of the graph topology (geographic vs. correlation-based) meaningfully affects the forecasting accuracy.

Section 2 defines the FL problem, features, and graph topologies. Section 3 describes the GTVMin objective and FedGD algorithm. Section 4 presents and discusses the results. Section 5 concludes.

2. PROBLEM FORMULATION

In the FL graph, each node is one FMI weather station. All stations with less than 80% hourly completeness over 2024 are excluded, leaving 165 stations. After feature construction, 29 stations that lack data for one or more non-wind variables are also excluded, leaving 136 stations to train the network on. The station distribution is also well divided geographically over all of Finland (fig. 1) so no geographical biases are introduced by this filtering step.

Our local models forecast the next-hour wind speed at station i . Formally, the prediction label is $y_t^{(i)} = ws_{t+1}^{(i)}$, the wind speed one hour ahead. The next-hour forecasts enable grid operators to schedule dispatchable backup generation and place intraday electricity marked bids before delivery. For features, the report uses lagged wind speeds at times $t-1, t-2, t-3$ and $t-1$ lagged atmospheric pressure, temperature and humidity:

$$\mathbf{x}_t^{(i)} = [ws_{t-3}^{(i)}, ws_{t-2}^{(i)}, ws_{t-1}^{(i)}, ta_{t-1}^{(i)}, rh_{t-1}^{(i)}, pa_{t-1}^{(i)}]^\top \in \mathbb{R}^6,$$

where ws (m/s) is wind speed, ta ($^{\circ}\text{C}$) air temperature, rh (%) relative humidity, and pa (hPa) atmospheric pressure. The lagged wind speeds are used because they are commonly used as predictors due to the temporal autocorrelation. In other words, consecutive time steps are correlated with each other. Pressure gradients are what create surface wind thus justifying it as a feature. Temperature and humidity don't directly cause wind, but they correlate with large scale weather patterns like storms that affect wind speeds.

Both features and labels are z-score normalized ($\mu = 0$, $\sigma = 1$) using only the training set statistics (so no validation or test data leaks into training). Feature normalization ensures that no variable dominates the model gradient due to its scale difference of physical units. Label normalization also removes each station's mean wind speed, preventing the bias term to encode geographic location. For example, with label normalization, two different stations from inland and sea-side might have similar wind patterns, but have different mean wind speeds. After normalizing, these stations can then theoretically communicate in GTVMin [4] (defined in section 3) and improve each other's models.

The local model for each station is a simple linear regression:

$$\hat{y}_t^{(i)} = (\mathbf{w}^{(i)})^T \tilde{\mathbf{x}}_t^{(i)}, \quad \mathbf{w}^{(i)} \in \mathbb{R}^7,$$

where $\tilde{\mathbf{x}}_t^{(i)} = [\mathbf{x}_t^{(i)T}, 1]^T$ is the feature vector augmented with a bias term.

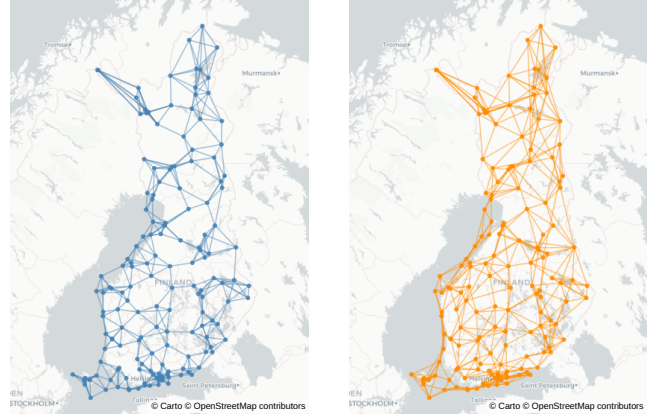
A linear model is chosen because only 6 features are used and each station has abundant training data ($\geq 80\%$ of 1 year hourly observations). Because of the massive amount of training data, no additional regularization is needed for the linear models. The closed-form gradient of the linear model also pairs well with the FedGD update rule.

The loss function is mean squared error (MSE):

$$L_i(\mathbf{w}^{(i)}) = \frac{1}{|\mathcal{D}^{(i)}|} \sum_{(\mathbf{x}, y) \in \mathcal{D}^{(i)}} \left((\mathbf{w}^{(i)})^T \mathbf{x} - y \right)^2 \quad (1)$$

MSE is chosen over alternatives such as mean absolute error (MAE) because MSE is differentiable everywhere. This property of MSE pairs well with gradient-based optimization and produces a clean and closed-form gradient for the linear models. The quadratic penalty also penalizes larger prediction errors more heavily, which is useful for capturing outliers in addition to typical measurements.

This report compares two methods for graph constructions. The first system is based on station geography. The first graph construction method connects each station to its $k = 5$ nearest neighbors. Connecting each station to its five closest neighbors creates a small local neighborhood that share similar characteristics. The distance between two stations is computed using the Haversine formula [5], which accounts for the curvature of the Earth.



(a) System 1: Geographic k-NN (b) System 2: Correlation k-NN

Fig. 1: FL network topology over Finland. Edges connect each station to its $k = 5$ nearest neighbours; edge colour indicates the system. Edge width is proportional to A_{ij} .

Edge weights in the Geographic k-NN system are assigned using a Gaussian kernel applied to the pairwise Haversine distances:

$$A_{ij} = \exp\left(\frac{-d_{ij}^2}{\sigma^2}\right) \quad (2)$$

where d_{ij} is the Haversine distance between stations i and j , and σ^2 is the median squared pairwise distance (a data-adaptive bandwidth).

System 2 connects each station to its $k = 5$ most correlated neighbours. The correlation is measured with Pearson correlation on 2024 wind speed series:

$$A_{ij} = \frac{\sum_{t=1}^T (y_t^{(i)} - \bar{y}^{(i)})(y_t^{(j)} - \bar{y}^{(j)})}{\sqrt{\sum_{t=1}^T (y_t^{(i)} - \bar{y}^{(i)})^2 \sum_{t=1}^T (y_t^{(j)} - \bar{y}^{(j)})^2}}.$$

Negative correlations are clipped so the code uses $A_{ij} = \max(0, A_{ij})$, since GTVMin [4] penalty requires non-negative edge weights. Even though the local linear models use six features for forecasting, the station correlation is computed only using wind speeds. This choice was made because stations that share similar wind speed behavior should have their models drawn closer to each other. This graph construction method should in theory be able to capture meteorological similarities beyond just geographical proximity.

3. METHODOLOGY

The graph total variation (GTV) [4] penalty uses the squared ℓ_2 norm

$$\|\mathbf{w}^{(i)} - \mathbf{w}^{(j)}\|^2 \quad (3)$$

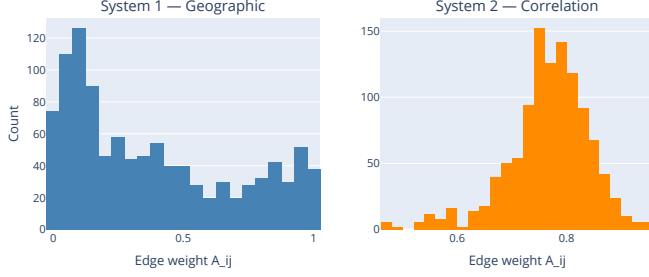


Fig. 2: Distribution of edge weights A_{ij} for System 1 (geographic, left) and System 2 (correlation, right).

to measure disagreement between neighboring stations' parameters. The squared ℓ_2 norm is chosen over an ℓ_1 -based penalty for three reasons. First, it is smooth and convex everywhere, giving a closed-form gradient that pairs directly with the FedGD update rule. Second, it penalizes large disagreements quadratically, encouraging tight parameter agreement rather than merely bounded spread. Third, adding a strictly convex penalty to the sum of convex local losses makes the full GTVMin objective strongly convex for $\alpha > 0$, guaranteeing a unique global minimizer.

With eq. (1) and eq. (3) the full GTVMin objective is

$$\min_{\mathbf{w}^{(1)}, \dots, \mathbf{w}^{(n)}} \underbrace{\sum_{i=1}^n L_i(\mathbf{w}^{(i)})}_{\text{local fit}} + \alpha \underbrace{\sum_{i,j \in \mathcal{E}} A_{ij} \|\mathbf{w}^{(i)} - \mathbf{w}^{(j)}\|^2}_{\text{GTV penalty}} \quad (4)$$

where α is the collaboration strength. If $\alpha = 0$, there is no penalty for differing models. If $\alpha = \infty$, all stations will share their model. So raising the value of α encourages collaboration and decreasing it reduces collaboration.

The GTVMin (eq. (4)) is solved with synchronous federated gradient descent (FedGD) [2]. Using FedGD makes every node update simultaneously. During each iteration, the weights are updated by

$$\mathbf{w}^{(i,t+1)} = \mathbf{w}^{(i,t)} - \eta \left[\underbrace{\nabla L_i(\mathbf{w}^{(i,t)})}_{\text{local gradient}} + 2\alpha \underbrace{\sum_{j \in \mathcal{N}(i)} A_{ij} (\mathbf{w}^{(i,t)} - \mathbf{w}^{(j,t)})}_{\text{weighted disagreement}} \right].$$

Each iteration, node i broadcasts $\mathbf{w}^{(i,t)}$ to its neighbours and receives theirs; all nodes then update simultaneously. The raw data $\mathcal{D}^{(i)}$ never leaves node i – only model parameters are shared.

The following hyperparameters are used: All models are initialized with weights drawn from $\mathcal{N}(0, 0.01)$ and trained for 750 iterations with learning rate $\eta = 0.01$. For the α -sweep, 300 iterations are used per candidate value.

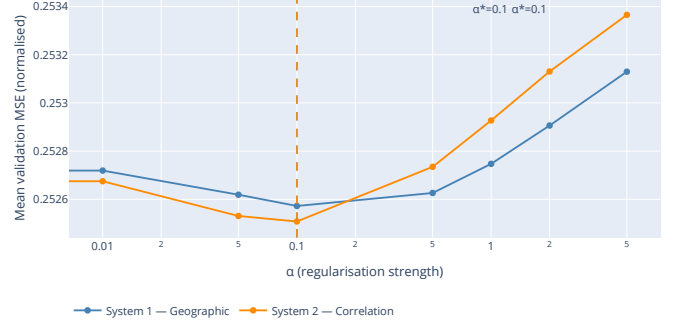


Fig. 3: Mean validation MSE vs. regularization strength α for both systems. Dashed lines mark the selected α^* .

4. NUMERICAL EXPERIMENTS

As data, the report uses FMI Open Data [1] hourly observations for the full calendar year 2024, fetched via the WFS `fmi::observations::weather::hourly::simple` stored query with a bounding box covering Finland (lat 59.5–70.2°N, lon 19.0–31.6°E). The four variables used are hourly mean wind speed (`ws_pt1h_avg`), air temperature (`ta_pt1h_avg`), relative humidity (`rh_pt1h_avg`), and atmospheric pressure (`pa_pt1h_avg`). Stations with fewer than 80% of the 8 784 possible hourly wind speed observations present are excluded, leaving 165 stations. After feature construction and dropping rows with remaining missing values, 136 stations have sufficient data and form the final FL graph. Small gaps of up to two consecutive hours are filled by linear interpolation before feature construction; larger gaps are left as missing and the affected rows dropped.

The data set is split into a training, validation and test split. With time series data, the split has to retain chronological order so that future data is not informed by past data. The report uses a 70/15/15 split: 70% of the chronologically first data points are in the train set, 15% in the validation and 15% in the test set.

Before comparing the performance of the two graph construction methods, an optimal hyperparameter α has to be chosen. This is done by running the FedGD algorithm for a fixed number of iterations and measuring the mean validation MSE across all stations. From fig. 3, both graphs perform the best when $\alpha = 0.1$. This value for α will then be used with both systems to compare their performance.

The baseline is the purely local model ($\alpha = 0$).

After running the FL experiments, the results are listed in table 1. The table shows that all systems perform basically identically. Looking at the test MSE, there is a marginal improvement, but this is practically an insignificant difference. Also when looking at the convergence graph in fig. 4, the systems show basically identical convergence behavior. The training loss also converges smoothly for all systems, which indicates that the linear model is a good choice for this appli-

Table 1: Mean MSE across all stations (normalised labels). Bold marks the best test MSE.

System	α	Train MSE	Val MSE	Test MSE
Baseline ($\alpha = 0$)	0.00	0.2345	0.2381	0.2638
System 1 — Geographic	0.10	0.2347	0.2377	0.2630
System 2 — Correlation	0.10	0.2349	0.2377	0.2628

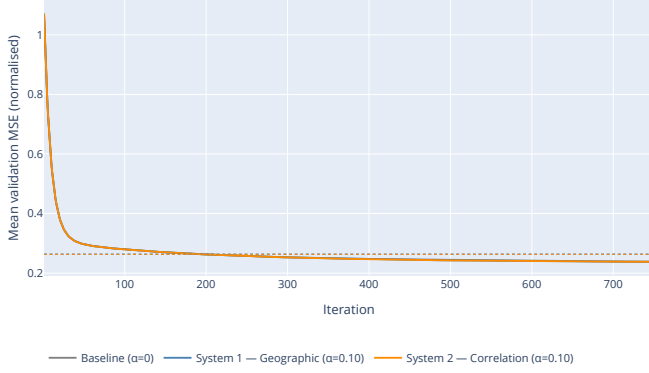


Fig. 4: Validation loss vs. iteration for both systems at their selected α^* , compared to the baseline ($\alpha = 0$). Dotted lines mark the final test RMSE.

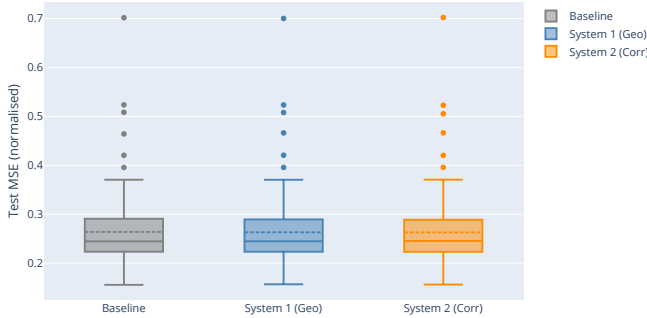


Fig. 5: Distribution of per-node test MSE (normalized) across all stations for the baseline, System 1, and System 2.

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The fig. 5 shows how the test MSE are distributed among all three models. The main insight from fig. 5 is that none of the models are able to reduce the test MSE of the outliers. This can be explained by the bigger variation of certain stations. The fig. 6 shows the best fit and the worst fit based on MSE. From the figure, there is a clear pattern that the worst fits are caused by more variation in the data, not the worse quality of the model.

5. CONCLUSION

The report tested whether federated learning over a geographic or correlation-based graph improves next-hour wind

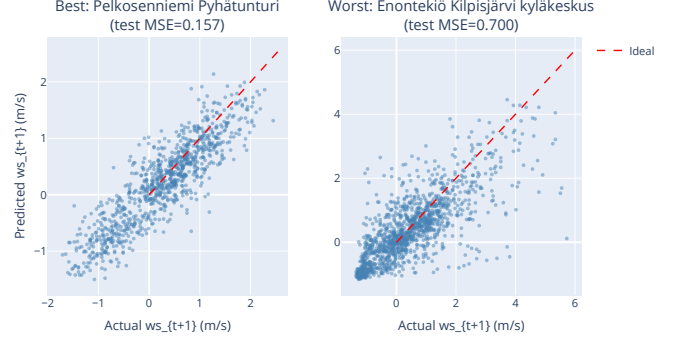


Fig. 6: Predicted vs. actual next-hour wind speed on the test set for System 1. Points near the diagonal indicate accurate predictions.

speed forecasting. The report used a purely local model as the baseline to research the effects of the federated models. When comparing the FL models to the baseline, the report found that both FL systems achieved negligible improvement over the baseline (table 1), indicating that collaboration provided no practical benefit in this setting.

The α -sweep selected a small regularization strength ($\alpha = 0.1$), which suggests that strong inter-station collaboration is harmful. The boxplot (fig. 5) confirms the aggregate finding at the per-node level: FL neither rescues the worst performing stations nor degrades the best performing ones. The negative result is because of the simplicity of the local model. A 7-parameter linear regression trained on abundant local data leaves little room for graph regularization to help.

Several directions could extend this work:

- **More expressive local models.** A shallow MLP or LSTM would introduce non-linearities a linear model cannot capture, giving graph regularization more room to improve performance.
- **Sparser data regimes.** With shorter records or higher missingness, borrowing from neighbours becomes more valuable.
- **Wind-flow-aware graph construction.** A directed graph based on prevailing wind directions could capture asymmetric weather-front propagation that symmetric k-NN graphs miss.

6. REFERENCES

- [1] Finnish Meteorological Institute. FMI open data. <https://en.ilmatieteenlaitos.fi/open-data>, 2024. Hourly weather observations accessed via WFS API. Referenced: May 17, 2026.
- [2] A. Jung. *Federated Learning: From Theory to Practice*. Springer, 2026. Arxiv Preprint: <https://arxiv.org/abs/2505.19183>.
- [3] Yang Li, Ruinong Wang, Yuanzheng Li, Meng Zhang, and Chao Long. Wind power forecasting considering data privacy protection: A federated deep reinforcement learning approach. *Applied Energy*, 329:120291, 2023.
- [4] Yasmin SarcheshmehPour, Yu Tian, Linli Zhang, and Alexander Jung. Clustered federated learning via generalized total variation minimization. *IEEE Transactions on Signal Processing*, 2023.
- [5] R. W. Sinnott. Virtues of the haversine. *Sky and Telescope*, 68(2):159, 1984.
- [6] Statistics Finland. Altogether 95 per cent of Finland’s electricity production was based on fossil-free energy in 2024. Official Statistics of Finland (OSF): Energy supply and consumption [online publication], April 2025. ISSN=1799-7976. Referenced: May 17, 2026.